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SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

TOPIC 7: Practices for Text-Based Requirement Documentation (Part I)

Innovating Solutions

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Contents

- Natural-Language-Based Work Products
  - Problems with Natural language
- Techniques to overcome problems with Natural language
  - Template-Based Work Products
  - Glossaries

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Type of Requirements Documentation

TEXT-BASED



Model-Based
Topic 4

Text-Based
- Natural language



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01 NATURAL-LANGUAGE-BASED WORK PRODUCTS

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NATURAL LANGUAGE

What

- A language that people use for speaking and writing in everyday life

Advantages

- Unconstrained natural language is extremely expressive and flexible
- Almost any requirements in any aspect can be expressed in natural language.
- No specific training is required to read and understand natural language.

Natural-Language-Based Work Products



Disadvantages

- Text written in natural language can frequently be interpreted in many way - ambiguities issue.
- Detection of ambiguities, omisions and inconsistencies is texts is difficult and expensive

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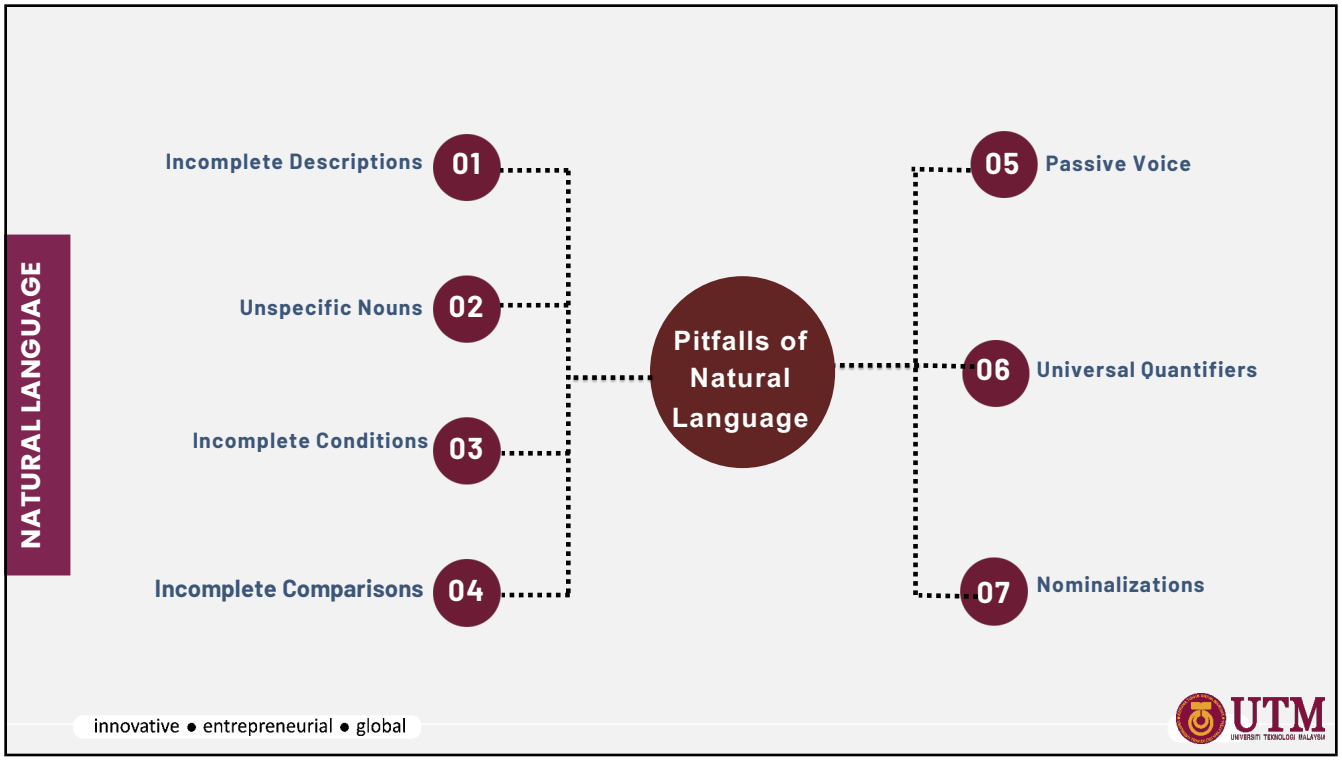
Natural-Language-Based Work Products

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Cause of Errors: Incomplete Descriptions

NATURAL LANGUAGE

- Some **verbs** require more than just a noun to be completely specified.
- Example Incomplete Requirement:
  - To log a user in, the login data is entered
- Why it is incomplete:
  - What login data is entered?
  - How is the data entered?
  - Who enters the data?
- Improved Requirement:
  - The user (**Who**) shall log in to the system by entering his username and password (**What**) by using the keyboard of terminal or scanning the qr code (**How**).

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Cause of Errors: Unspecific Nouns

NATURAL LANGUAGE

- Vaguely specified nouns.
- Example Unspecified Nouns Requirement:
  - The data shall be displayed to the user on the screen.
- Why it is an error:
  - Which data is exactly?
  - Which user is exactly?
  - Which screen?
- Improved Requirement:
  - The system shall display the calculated data to the registered user on the User Analysis screen.

Cause of Errors: Incomplete Conditions

NATURAL LANGUAGE

- People focus on the normal case, omitting exceptional cases.
- Example Incomplete Conditions Requirement:
  - The restaurant shall offer discounts of 10% to registered user under the age of 20.
- Why it is an error:
  - What about age more then 20 inclusive?
  - Need to consider all cases - the else case
- Improved Requirement:
  - The restaurant shall offer discounts of 10% to registered user under the age of 20.
  - The restaurant shall offer no discounts to registered user over the age of 21.

Cause of Errors: Incomplete Comparison

NATURAL LANGUAGE

- A requirement includes a comparison (e.g., faster, better, higher) but lacks a clear baseline, reference object, or measurable criteria for the comparison
- Example Incomplete Comparison Requirement:
  - The system must perform faster than before
- Why it is an error:
  - lacks a specific baseline (e.g., previous version's response time in seconds) or measurable metric (e.g., 20% reduction in latency), making it impossible to verify or implement consistently.
- Improved Requirement:
  - The system's average response time must be reduced to under 2 seconds, compared to the current version's 5-second average.

Cause of Errors: Passive Voice

NATURAL LANGUAGE

- Sentence in passive voice has no subjects.
- Example Passive Voice Requirement:
  - Notifications are sent when errors are detected
- Why it is an error:
  - The passive voice "are sent" and "are detected" omits the subject (e.g., which system component sends notifications or detects errors), making it unclear who or what performs these actions, potentially causing missed or redundant notifications.
- Improved Requirement:
  - The error monitoring service shall send notifications to admins when it detects critical errors

Cause of Errors: Universal Quantifier

NATURAL LANGUAGE

- Universal quantifiers in natural language requirements, such as "all," "every," "always," or "any," can cause errors when they imply absolute or overly broad conditions without considering exceptions, constraints, or specific contexts.
  - Not all the objects should be treated the same way.
  - Avoid: Each, every, never, always, each time, in the evening, all, some, nothing, none
- Example Universal Quantifier Requirement:
  - All users can access the system.
- Why it is an error:
  - The quantifier "all users" implies every possible user (e.g., guests, admins, unauthenticated users) without specifying roles or conditions (e.g., authentication, permissions), potentially leading to security vulnerabilities or incorrect access control.
- Improved Requirement:
  - Authenticated users with admin or editor roles shall access the system via the web interface after successful login

Cause of Errors: Nominalization

NATURAL LANGUAGE

- Verbs or actions are turned into nouns, often making the requirement less clear, more abstract, or harder to interpret.
- A long process is transformed into a single event. All information necessary to accurately describe the process is thereby lost.
- Example Nominalized Requirement:
  - The generation of reports happens daily
- Why it is an error:
  - "Generation" hides the action (generate) and the responsible entity (e.g., reporting module), leading to confusion about which component creates reports or what report content is required.
- Improved Requirement:
  - The reporting module shall generate daily sales reports in PDF format